

Homework-5

1. Let $X^n - a \in k[x]$ be irreducible/ k (a a field) and let b be a root of $X^n - a$ in some extⁿ K/k . Let $m \in \mathbb{N}$ be a divisor of n . Find the degree of $\text{Irr}(b^m; k)$.
2. Let R be an integral domain containing a field F such that $\dim_F R < \infty$, R treated as a vector space over F . Prove R is a field.
3. Find conditions on a and b such that the splitting field of $X^3 + ax + b \in \mathbb{Q}[X]$ has degree 3 over $\mathbb{Q}$.
4. For $f(x) = x^4 - 2x^2 - 2 \in \mathbb{Q}[x]$, find roots α, β such that $\mathbb{Q}(\alpha) \cong \mathbb{Q}(\beta)$. What is the splitting field of $f(x)$ over $\mathbb{Q}$?
5. Let α be a root of $X^3 - X - 1 \in \mathbb{Q}[X]$ and $K = \mathbb{Q}(\alpha)$. Prove that K cannot be the splitting field over $\mathbb{Q}$ of any polynomial/ $\mathbb{Q}$.
6. Let k be a field, $\text{char}(k) = 0$; $f(x) \in k[x]$ be a polynomial of degree n ; $a, h \in k$. Prove that
$$f(a+h) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a) h^k, \quad \text{Where}$$
$$f^{(k)}(x) = (f^{(k-1)}(x))', \quad f^{(1)}(x) = f'(x).$$

7. Let L/k be the splitting field of $f(x) \in k[x]$. Let $K \supseteq k$ be a subfield of L . Prove that any k -embedding of K into L can be extended to an automorphism of L .
8. Let L/k be a finite extension, $[L:k] = n$. Let E/k be any field extension. Prove that the number of distinct k -embeddings of L into E is at most n .
9. Let $f(x) \in k[x]$ be a monic polynomial with distinct roots $\alpha_1, \dots, \alpha_n$, $\deg f = n$. Prove
- $$f'(x) = \sum_{j=1}^{n-1} (x-\alpha_1) \dots (x-\alpha_{j-1}) (x-\alpha_{j+1}) \dots (x-\alpha_n).$$
10. Let k be a field of positive characteristic p . Prove, for $a, b \in k$, $(a+b)^p = a^p + b^p$. Let $\mathbb{F}$ be a finite field with $q = p^n$ elements. Prove that the map $\varphi: \mathbb{F} \rightarrow \mathbb{F}$, $x \mapsto x^p$ is an automorphism, $\varphi^n = \text{Identity}$.
11. Let $f(x) \in k[x]$ be monic of positive degree. Prove that all the roots of f in a splitting field are simple (i.e. have multiplicity 1) if and only if $\gcd(f, f') = 1$.